ActInf Textbook Group ~ Cohort 7 ~ Session 1 (Chapter 1, Part 1) ~ 7/15/20

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all right hello cohort 7 thank you
everyone for all the introductions and
and saying
hello this is our first meeting so we've
just all introduce
ourselves and now we'll talk a little
bit about chapter one
slash anything active inference related
kind of funneling into the textbook
anything about the positioning of the
textbook overall the two years since the
publication and any topics that people
are like excited about to explore in
that third week which will give us like
a little rest from the Rhythm otherwise
um but let's go to chapter one and just
like where does anyone want to begin
what's a any whether it's a quote or
section from chapter one or it's just
something about their approach to active
inference or it's a it's a general
question uh that they
have where where would anyone like to
begin I'd like to ask a oh yes please
please yeah so um um there's a statement
on page
five saying that active
evence as a normative approach to
understand brain and mind can you
explain to me what normative approach
means great great question I was
wondering the same thing yeah yeah this
is one of this is a this is a very
important
question I I I'm just going to where I I

knew that the question had been asked
but um Ali or anyone does anyone want to
give a thought what what does normative
mean when we're talking about active
inference and a normative theoretical
perspective yes
ali uh well yeah there's an explanation
for the use of this term normative in
the notes and specifically on page um
267 I guess uh so the use of this term
uh differs a little bit from its normal
usage for example in philosophy
literature or some in some other
discourse uh so here the term normative
basically refers to uh the way active
inference
modeling um
somehow uh works works according to some
evaluative standards so uh more
precisely uh in active inference we have
uh a specific parameter say surprisal or
free energy which uh will get familiar
uh more extensively in the uh coming
chapters but uh there's this parameter
around which all the other elements are
evaluated uh and manipulated and modeled
and so on so the word normative here
actually refers to uh the framework of
uh this Theory um and um it it's
not uh this is at least it's not exactly
used in the same meaning as used in for
example uh in ethics or in some I don't
know in other usage it's used in a in a
rather technical way here so yeah thank
you Ali yeah I added a link here to the
social sciences course from last year
and AVL and Ma work and Ben and ren too
is discussing the social normativity

social constraints and so that kind of
like ought versus is and like what one
ought or normed to
do as Ali mentioned mentioned um but but
it but
it's we've explored that from some other
directions too like to what extent does
this outline how one ought to model
Michael sure um yeah it is moving on to
a different subject
[Music]
um uh I was kind of
wondering uh
so the agents or entities uh often have
like a sort of highest preference with
prior um I'm wondering if
theoretically Andor practically I
suppose these policies are always kind
of
externally artificially injected as like
sort of vacines or something like uh
Fine Food
say or if Act of inference is I guess I
don't know if the word self-consistent
um but if it these priors
manifest
automatically um by the I suppose
perhaps uh notion of minimizing
surprise or the uh the the bias priors
are artificially injected every
time yeah this is a this is a great
question anyone have a thought on on
it well
normative even in this discussion is
sometimes used in a very strong sense
the core body temperature of a mammal
has to be between a certain minimum and
a certain maximum or that animal will

stop moving that animal will die you
have to have a certain amount of
oxygen or you will stop
moving and die
uh there are some less violent
impositions for instance anyone walking
in public in a US city has to wear
clothes and if you don't you will be
taken away and therefore you will not be
walking in a city
anymore so then you can get into much
more optional things like people have to
be minimally nice to one another or
people will stay away from you but uh
there's a great range of uses uh all of
which
have the sort of ordinary language sense
of ought to or have to must must not I
think any of those things
are norms of one kind or another and can
be thought of
normatively cool yeah I guess um you
know they kind of help Kickstart and
bootstrap the framing of a problem of
Interest anyways otherwise how do you
you have to Define boundaries at some
point I suppose
yeah yes the system of Interest choice
and the boundary identification question
in theory and practice this will come up
a lot um Ali and then anyone else who
raise their
hand well yeah regarding Michael's
question about uh how exactly prior SL
preferences
are implemented in the agent well I
believe um it's both I mean it it can be
implemented uh purely

artificially in in an artificial way or
it can arise from um as in activist
would call through sense making uh so uh
it can be pretty autonomous uh and uh
it's not required uh to somehow
implement those priors and
preferences uh through some external
implementation uh in self organ in
self-organized agents and
specifically in the organisms uh that
participate in the sense making uh I I
use this term uh in the inactive um
inactive sense uh those priors and
preferences can arise from
uh the very active sense making uh so
yeah it can
be I mean purely internal and autonomous
as
well
thanks cool yeah there's so much to say
on this like in the textbook we're
learning about some of the Lego pieces
and the the building blocks and some of
the names of the colors and like that
level of of fundamental and then we work
in the second half of the textbook on
some simple examples like listening to
music and anticipation a rat in a tze
some other kinds of of simple mapped
cases like a pendulum swinging in
physics and then the
sophistication and the the complexity
hierarchy is essentially open-ended
because this is a cognitive modeling
approach or systems modeling approach so
when we're talking about complex
territories then it's like a complex map
so a lot of times there's

this movements or tendency to to and and
and an effective one at that to kind of
like ground in a more basic example or
Motif like something that we'll see in
the textbook but then when we look at
like a complex system like an organism
or social setting then that's that's Str
stretching the limits of our like
mapping and modeling
capacities and then we can kind of
supplement that movement from the
simpler Motif to the more complex system
of Interest supplement that but okay
well what what what is this new recent
Ryan Smith paper on interoception like
how are they modeling it here and so
that helps enter in like more modern and
empirical work
otherwise we can go very simple look at
the sort of
Essentials that's uh and then we can
also take it to um strive to model more
complex
systems any question or thought we can
just or we can also look at some of
the top prior questions
so one feature that comes up in in um
chapter one well there's two figures in
chapter
one this is the diotic phrasing of agent
and Niche there's a deeper symmetry so
it could be like agent agent Niche Niche
it's just often represented as like a
mobile agent in a world however the
framework for interfaces boundaries
blankets is more generic than that but
this is kind of the two-part model with
observation coming in action going out

and then that is going to get nuanced into more of a four-part model where there's two things coming in and then there's two things coming out but we'll we'll get there so this is sort of just situating active inference within this ecological psychological tradition the cybernetic of agent and Niche and then we have the famous figure 1.2 which introduces and summarizes the low road and the high road this is a really important distinction that was introduced lightly in some earlier work it's referenced in a few of the questions uh about the desert landscape and here we have active inference as it's kind of like a metro map it's a schematic metro map so it's not like these are the only and the only ordering of terms along the way so don't commute with this map per se but it does describe the low road and the high road so-called so does anyone just like want to give any thought or overview how did they see the the low road and the high road what distinguishes them how does this concept like come into play in the textbook um I don't know if this is a d point but I guess what interests me is somebody who doesn't know very much about bism at all but would like to is that it's interesting that they are too separate directions I guess naively I would have thought that the free energy

principle was quite reliant on base
theorem because of the definition of
surprise so I guess it's interesting to
me the idea that there are
two roads and one doesn't kind of depend
on the other but they meet at one
place interesting thank you thean and
then
Frasier I I've looked into
this I'm I'll make a comment but it
might be a question um so my sort of
take on things is that the low road
is a perceptually
based and that's distinctly different
from the free energy principle side
through predictive processing into
active
inference that is
about um survival and resisting entropy
and and that line of things so is that
correct yeah that's that's great we well
let's revisit this
Fraser yeah just a thought on the low
road and the high road to my mind I mean
there was that great 15minute video of
Carl fron talking about the low road and
the high road um as far as I can recall
the the low road is perhaps you know
we're starting with an idea about very
very small uh micro states of some
system and we're going to perhaps maybe
this is a gas or something and we want
to have a look at how those micro states
are interacting and then through a kind
of amalgamation of this you know the
behavior of these micro States we can
maybe get out of that an abstraction
something like adaptive Behavior whereas

the the higher road is a bit more like
okay let's just take a bit more of a
sort of an armchair approach and kind of
you know conceptualize from a top down
idea about what it would mean to be
adaptive at all um kind of starting from
an aggregate idea and then kind of going
down and saying okay well we have this
aggregate idea how are the micro States
then going to have to behave so that
that's kind of my thoughts on the the
low road and high road distinction the
only other thing I would like to mention
is on that that initial figure you
showed the agent observation loop I have
in in various other uh parts of the
literature cybernetics um cognitive
science and so on heard this referred to
as the the sensor motor loop I don't
know if they actually used that
terminology in the book I could be wrong
it has been a little bit since I've
actually read the book but that that
seems to be a relatively Universal way
to talk about this process of cyclical
interaction between a system and its
environment so I hope that's somewhat
interesting thank
you
yeah so there's there's a lot of ways
to talk about the in the high road um
does anyone else want to just like give
was any other thought on on low or
high a textbook scale point to make is
this sets up chapters two and three
chapter two is going to take us from the
low road to active inference chapter
three is going to take us from the high

road to active inference and then chapter four is like the big like here you are here's the ACT inference generative model this is what you write in RX infer this is what you write on the napkin this is what you're sketching in the diagram like that's the active inference generative model which is both constructed along the low road path and compliant under the free energy principle umbrella so that's why the active inference generative model is sort of like the meeting point or the overlap because it it's both a manifest artifact which is what allies it with these kinds of model architectures found all along the low road and it's a modeling output artifact which is compliant with this free energy principle and the basy mechanics so that's why sometimes people will talk about active inference as well it's a Cory it's like a downstream or it's a derivative of the free energy principle so it's like this is how you take free energy principle which we'll go into and apply that to persistent systems or adaptive or dynamic systems or taking the low road like cobbling and tinkering together like if you start with Bas theorem now the last name does basically no work here base theorem is just about conditional probabilities for example between observations of data and abductive inferences to to latent States but we have here like if you wanted to build an architecture for perception and action

even the special case like just a
perceiver or just an actor if you wanted
to do that in a fully synthetic setting
like in silico like building agentic AI
or building an API call or something
like that or if you wanted to use this
more of like an empirical cartography
sense like make making a model of a
sensory motor system you're going to
start implementing these different kinds
of modeling choices and so built things
are all along the low road which looks a
lot in many cases like machine learning
artificial intelligence these neural
networks linear algebra stochastic
dynamical processes and building up
towards artifacts that contain this
unified approach to perception action
and one note on that coming only from
the low road here not bringing in fvp
per se is in Signal processing which is
dealing with that inbound perception
question action is often neglected and
then on the contrary in control theory
like balancing the plate on the stick
and Mountain car all that in control
theory the perceptual question is often
neglected and so one piece of the low
road approach here is taking a unifying
approach to perception action not
erasing their differences but having
them as part of a unified imperative
evidence maximization surprisal
minimization bounded by the variational
free energy so that's kind of one
element is having perception and action
in a unified model and then kind of a
broader commitment to the unifying of

all these diverse cognitive phenomena
like attention memory learning
anticipation all these different kinds
of phenomena which otherwise it would be
totally grab bag H how would these be
linked and there could be like infinite
pairwise combinations and models that
might not survive having a third piece
of the puzzle added but by taking this
kind of broader unificationist
perspective the the whole iguana
perspective like we're going to use an
integrated toolkit to model the whole
thing in its phenomena as it appears to
the modeler that's what gives Unity on
the low road and then that's without
getting into the free energy principle
and and what it brings from that first
principes perspective
theen oh
sorry or anyone else who wants to raise
their hand or or give a
thought um yeah I just add one thing to
your your comment on the the grab bag of
like attention and memory and everything
like that because it's actually
something we're working on within the
marketing field right now um could you
speak more to like
how this this approach would help keep
that from being that grab bag because
that is basically the way it works in a
practical sense at least in my industry
yeah great question what does it really
look like or mean for active inference
to be a unifying approach to modeling
different cognitive phenomena what would
anyone

say yeah Fraser
hi oh I'll let uh someone else go I've
talked already I heard
someone sorry that was me
did did you want to go Tom
or uh well I I I guess um and this so
I'm sure you'll have a smarter thing to
say but um I guess my understanding of a
lot of these sort of predictive
processing models was that there's no
difference really between a sort of a
desire and a prediction and so the sort
of this sort of classical psychology
state where you have a want and you act
on the world and you know what your
Desir is like and you know when you've
kind of reached that goal because you're
like what you reserve the reality
matches what your prediction was is sort
of collapsed so that free energy sort of
replaces like through surprisal replaces
this old distinction between like what a
desire is and what a prediction about
the world is so my understanding is that
those are united there and the attention
is seen more as a sort of
I guess a process by which that's
achieved as in when you attend to
something you're reting your prior about
it but I could be wrong there um that
could all be nonsense
but yeah frer thank you Tom I mean yeah
that that is in essence what I was going
to to recapitulate I mean the so the
initial question was how might active
infants bring a sort of un unifying
approach to cognitive phenomena in
general I think that was the question um

I I'm a very very big fan of the work of
uh Dr John V at the University of
Toronto is's a cognitive scientist so
his kind of whole Obsession in life is
this idea of how it is that um we
maintain adaptivity per se as an agent
what makes us intelligent and he thinks
it's this this process called relevance
realization we're able to zero in on the
relevant information essentially um so
all cognitive processes under this idea
would have to do with ability to zero in
on the relevant information and what
what might that be well in this case we
have this framing of a sensor motor Loop
you know this econ Niche that we're
constructing for ourselves to maintain
our our integrity across time so I guess
in in the in the broadest possible
conceptualization you know active INF is
a way of talking about
how the adaptivity of that sensor motor
Loop is is tuned uh in order to result
in the perpetuation of the sensor motor
Loop so and what you have to do what
that boils down to is what Tom said
basically is you have to have a you have
to embody a model of what your world is
in order to be able to uh match and and
be in sync with it so that you can you
can avoid things that are dous to your
your boundaries and you can you can take
in things that are good for you and so
on so I'll stop
talking he yeah there there's a lot to
say and I and I I think we'll also be
able to pick out some uh things that the
that the authors the textbook say I'll

give kind of a a a just a a a primer
response on this which is to go to
figure 4.3 so we're looking ahead to
what an active inference generative
model can look
like in this model whether looking at it
in the discrete time or in the
continuous time
setting there aren't like advanced
cogn there isn't
self-reflexivity there's not like
material reconstruction so there are
phenomena in the real world in the real
territory that are not in every single
active inference map part of the cool
thing about active inference is you can
apply it to a rock not that it's the
best tool but you could you could apply
it to um a cloud as it kind of coalesces
over a mountain or something like with a
rain shadow so it doesn't need to apply
to Any Given system or cognitive system
and so because of that generality one
way to think of it is active inference
says less as a technique which enables
people to say more and different about
systems of Interest so how does this
relate to how different psychological
phenomena cognitive phenomena get
unified in practice so let's just say
that somebody says well I'm interested
in the relation ship between sensory
Precision prior beliefs and agency and
estimates of efficacy in the world well
without a first principal's approach
that could be just a mega open-ended
literature search you're going to hit on
every single paper and journal on all

those and then you know
what another way would be to say okay
well I'm going to associate sensory
Precision with the a matrix here I'm
going to associate prior beliefs and
their possible differences with d and
with efficacy I'll associated with b and
I'll do this and this and that and so
I'll kind of compose my Legos together
and then I'll do model sweeps to explore
the accuracy minus complexity of model
fit like model preference and
efficacy for this empirical data set
which one of these synthetic models fits
the data best so or maybe we just want
to do the big one or or and test
something specific about the biggest
possible combinatoric model so even
though this can initially feel like less
of an answer what enables it to actually
be more of an answer through time is
that active inference isn't opinionated
on what it would even mean for attention
to connect to agency but then in
practice when you see a paper says by
agency we mean this by attention we mean
this and here's how we modeled their
interactions so that's the unified
compliant
playground that allows modelers to
actually specify and then have
interoperability under the unified
imperative so then models can be tested
like with Factor one factor two Factor
one and two and then there's actually a
coherent way to
say where one is more preferable and
what data set they have what kind of

performance on instead of getting into
these like more ontological
debates about you know well how does it
happen in the
brain well then that's great to zoom in
and study how it really might be
happening in the brain but that will
never give a generic answer because now
we're looking at a system of Interest
Tom thank you this is a very basic
question um so this idea is like you
know you might have like one Matrix that
means attention or it mathematically
instantiates this idea does that mean
you could create an
organism with some capabilities like
some kind of cognitive capabilities in
the model that don't exist in the real
world you know so you're not you don't
say as a framework you need to have some
mathematical bit that does attention
some mathematical bit that does
prediction but it that it's generic
enough that you could instantiate really
any kind of thing that you could receive
as a cognitive
capacity yeah that's a great way to put
it um one thing that Maxwell ramstead
said in the Discord that I remembered a
few weeks ago was the focus on agents
has led people to underestimate the
scope of free energy principle basy
mechanics so not just is AC of inference
unopinionated on cognitive architectures
of Any Given agent it's unopinionated on
how one might distinguish or model even
or even need there to be agents you
could just make a fabric of API calls or

you could make any or you could make a
fantasy subway map that doesn't need to
have any resemblance to any known
cognitive system or any extance system
at all again will that be useful for
this or that will somebody pay you for
this or that will that startup work I
mean those are secondary questions but
when we see that well we're talking
about like base graphs and then we're
going to associate incoming
directionality with perception and
outgoing with action but that doesn't
need to have any kind
of that's that's I'm not going to say
it's the limits of what can be said
there but that is the
toolkit and then someone can make an
architecture that has any kind of
existence within an arbitrary Niche
especially if it's in an immortal
setting like the Mortal Computing that's
where it gets into the the strange Loop
and the embodied and the survival and
and that's a very interesting Direction
but especially in silico where the
survival imperative is just the
modeler's
Whimsy you can make arbitrary things and
making simple pedagogical things is
often a a totally critical step like
when we look at the the music modeling
case in chapter
7 this this isn't claiming to a total
model of
Aesthetics or of culture someone says
yeah but also they they might have to
breathe where's the oxygen in this model

it's like it's it's just one facet or
Motif that makes sense thank
you yeah it's really fun to to explore
like what are the limits and the
constraints of modeling of
formalization of experimentation
and then wi within that how does active
inference differ from near relative
approaches so there's like kind of like
a continent scale and then a more fine
scale differentiation sometimes like are
we comparing active
inference with some unnamed or names
qualitative
approach which is a little bit like an
apple and an orange comparison which
isn't necessarily a bad thing just
they're two different kinds of stances
or are we comparing active inference to
a model where perception and action are
not considered with a unified imperative
but then let's see what that model is
and then that's more of a model tomodel
discussion so be the difference between
two competing hypotheses or two
alternative models on a data set versus
the question that's broader which is
like should we model this data set at
all cool
well I hope this is a little flavor of
chapter one and and some fun people who
we'll be spending the coming months with
again if you want to join the 13
UTC we'll be in chapter 6 which is super
fun and a great starting point as well
which is the recipe and that's the steps
that we'll really go through to make
active inference

models in cohort 6 um or otherwise we'll
catch up and uh
Andrew
Andor Ali or maybe others will be
facilitating but um in the time between
people can add questions add and augment
just if you want to do something and
contribute to it and you don't know what
again just let blanket
know okay cool thank you till next time
thanks everyone bye thank you everyone
thanks everyone